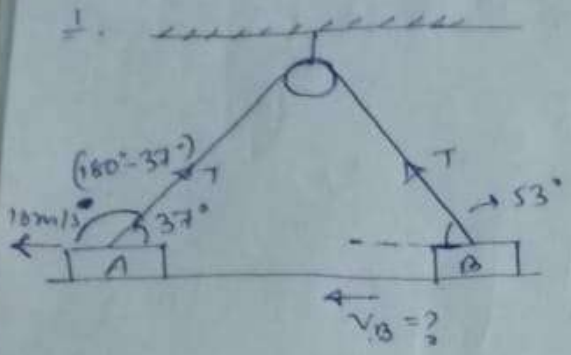
Constraint Motion (JEE) Solutions

3, 4, 5, 15, 17, 18 done in video class.

1.



$\Sigma \vec{T}\cdot\vec{V} = 0$

$\cancel{T} \times 10 \times \cos(180°-37°) + \cancel{T} \times V_B \times \cos 53° = 0$

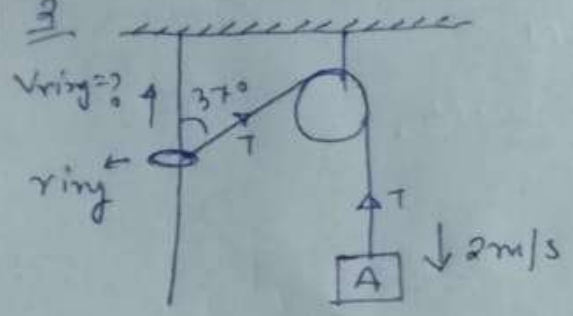
$-10\cos 37° + V_B \cos 53° = 0$

$V_B \cos 53° = 10 \cos 37°$

$V_B \times \frac{3}{\cancel{5}} = 10 \times \frac{4}{\cancel{5}}$

$V_B = \frac{40}{3} = 13.33 \text{ m/s}.$

3.



$\Sigma \vec{T}\cdot\vec{V} = 0$

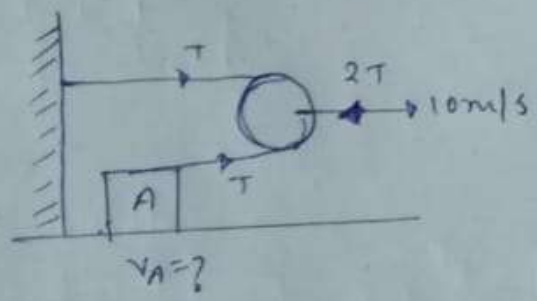
$-\cancel{T} \times 2 + \cancel{T} \times V_{ring} \times \cos 37° = 0$

$V_{ring} \times \cos 37° = 2$

$V_{ring} \times \frac{4}{5} = 2$

$\therefore V_{ring} = \frac{10}{4} = 2.5 \text{ m/s}.$

5.

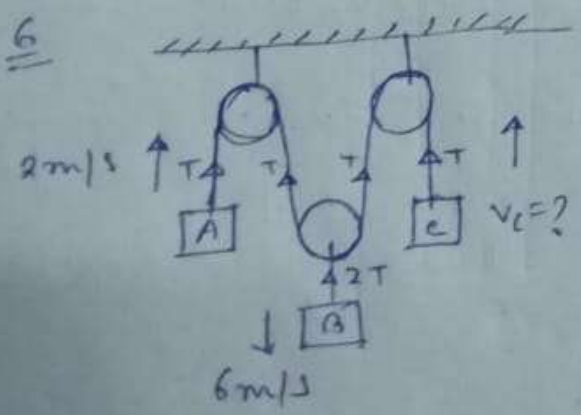


$\Sigma \vec{T}\cdot\vec{V} = 0$

$-2\cancel{T} \times 10 + \cancel{T} \times V_A = 0$

$V_A = 20 \text{ m/s}.$

6



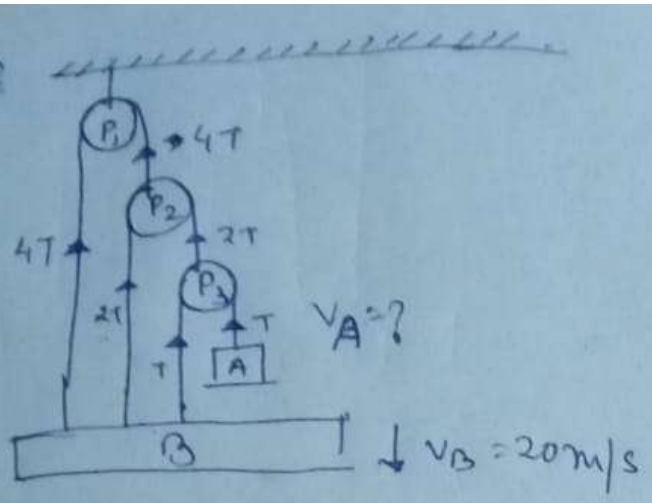
$\Sigma \vec{T}\cdot\vec{V} = 0$

$\cancel{T} \times 2 - 2\cancel{T} \times 6 + \cancel{T} \times V_C = 0$

$2 - 12 + V_C = 0$

$V_C = 10 \text{ m/s}, \text{ up} \uparrow$

16



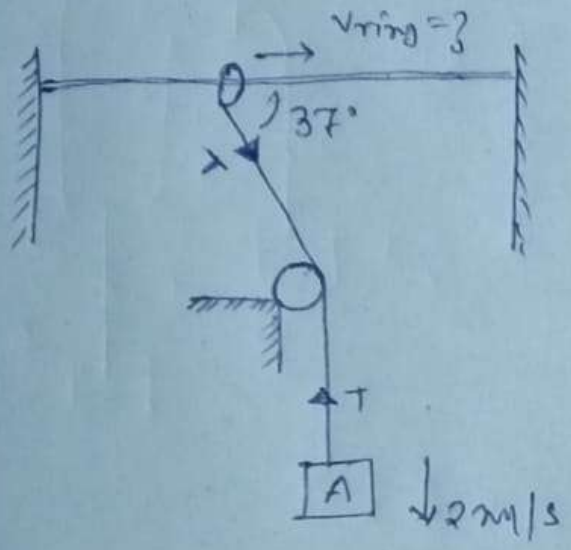
$$\sum \vec{T}\cdot\vec{v} = 0$$

$$-7\cancel{T}\times V_B + \cancel{T}\times V_A = 0$$

$$V_A = 7\times 20$$

$$V_A = 140 m/s, \text{ upward} \uparrow$$

19



$$\sum \vec{T}\cdot\vec{v} = 0$$

$$-\cancel{T}\times 2 + \cancel{T}\times V_{ring}\times \cos 37^\circ = 0$$

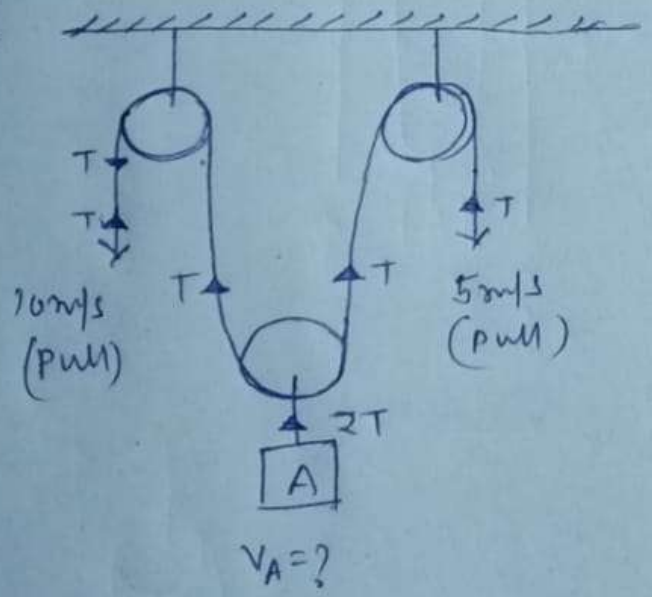
$$V_{ring}\times \cos 37^\circ = 2$$

$$V_{ring}\times \frac{4}{5} = 2$$

$$V_{ring} = \frac{10}{4}$$

$$V_{ring} = 2.5 m/s$$

20



$$\sum \vec{T}\cdot\vec{v} = 0$$

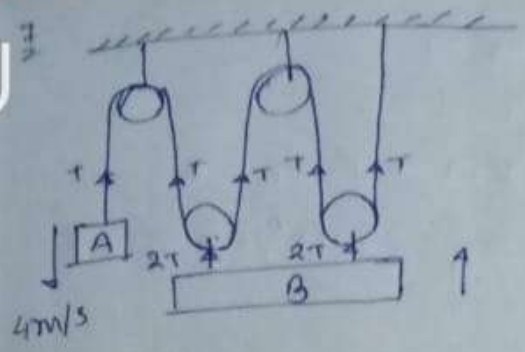
$$-\cancel{T}\times 10 - \cancel{T}\times 5 + 2\cancel{T}\times V_A = 0$$

$$-10 - 5 + 2V_A = 0$$

$$2V_A = 15$$

$$V_A = 7.5 m/s, \text{ upward} \uparrow$$

7



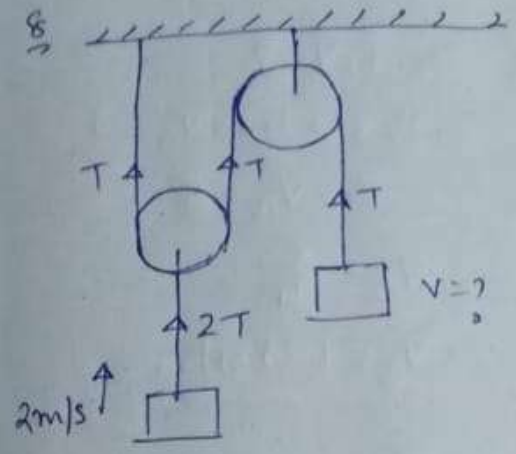
$\Sigma \vec{T}.\vec{v}$

$-\cancel{T} \times 4 + 4\cancel{T} \cdot v_B = 0$

$4 v_B = 4$

$\therefore v_B = 1\,m/s$, up ↑

8



$\Sigma \vec{T}.\vec{v} = 0$

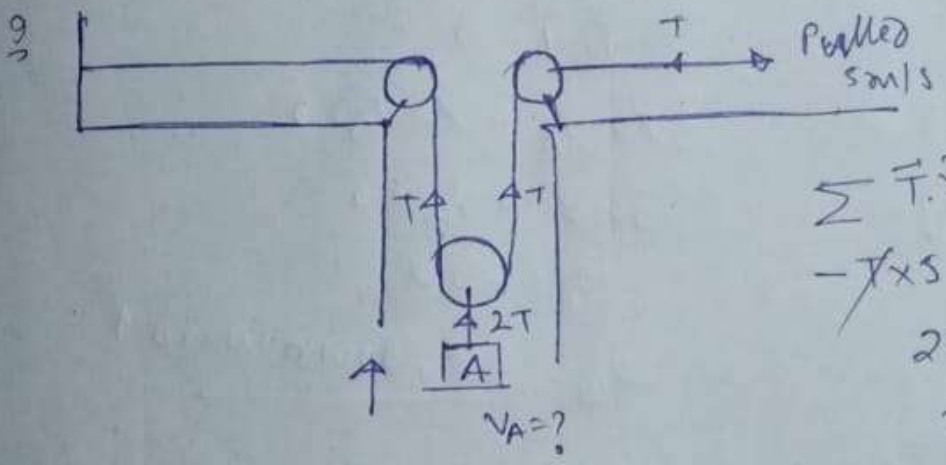
$2\cancel{T} \times 2 + \cancel{T} \times V = 0$

$4 + V = 0$

$V = -4$

$\therefore V = 4\,m/s$, downward ↓

9



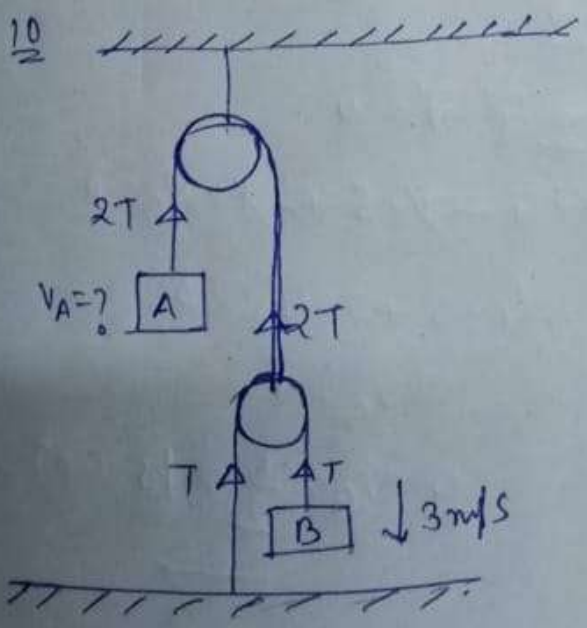
$\Sigma \vec{T}.\vec{v} = 0$

$-\cancel{T} \times 5 + 2\cancel{T} V_A = 0$

$2 V_A = 5$

$V_A = 2.5\,m/s$, upward ↑

10



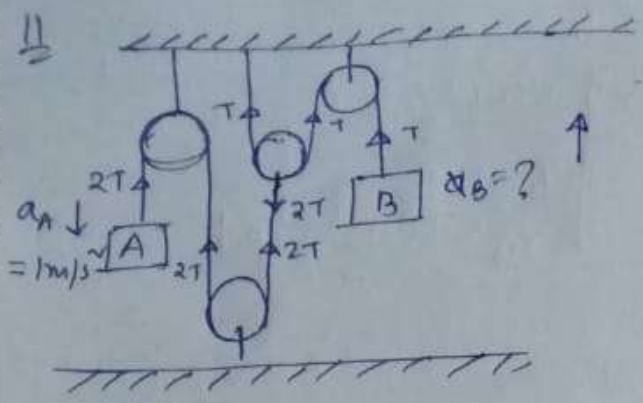
$\Sigma \vec{T}.\vec{v} = 0$

$-\cancel{T} \times 3 + 2\cancel{T} \times V_A = 0$

$2 \times V_A = 3$

$V_A = 1.5\,m/s$, upward ↑

11

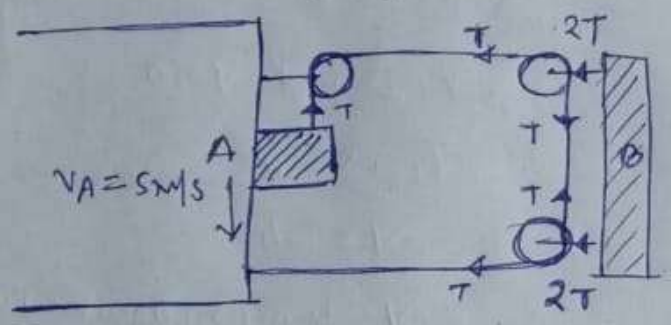


$\sum \vec{T} \cdot \vec{a} = 0$

$-2\cancel{T} \times 1 + \cancel{T} \times a_B = 0$

$a_B = 2 m/s^2$, upward ↑

12



$\sum \vec{T} \cdot \vec{V} = 0$

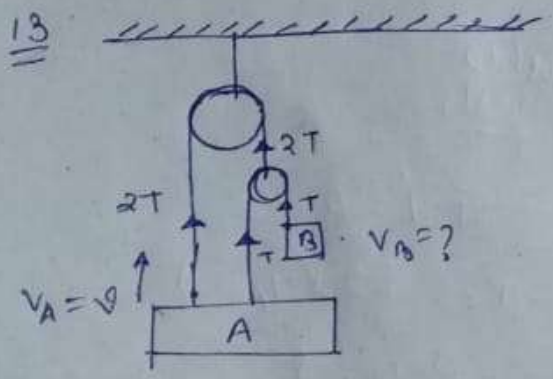
$-\cancel{T} \times 5 + 4\cancel{T} \times V_B = 0$

$4 \times V_B = 5$

$V_B = \frac{5}{4}$

$V_B = 1.25 m/s$,

13



$\sum \vec{T} \cdot \vec{v} = 0$

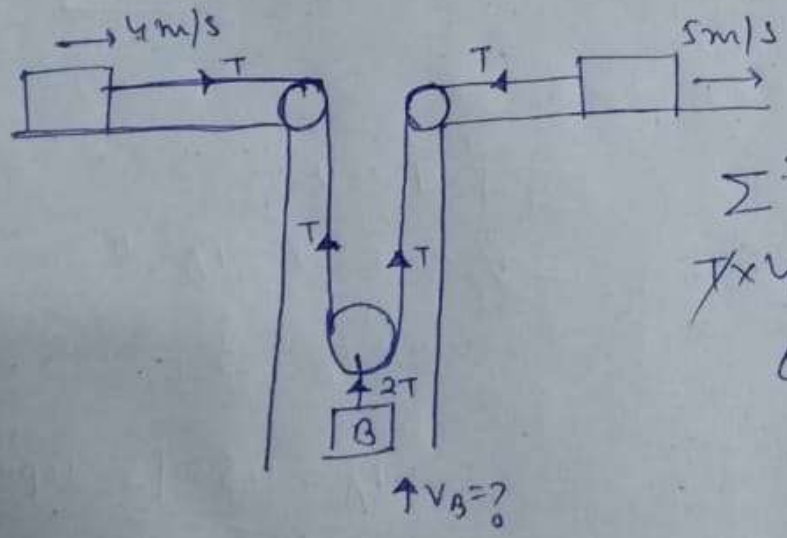
$3\cancel{T} \times v_A + \cancel{T} \times v_B = 0$

$3 \times v + v_B = 0$

$v_B = -3v$

$v_B = 3v$, downward ↑

14



$\sum \vec{T} \cdot \vec{v} = 0$

$\cancel{T} \times 4 - \cancel{T} \times 5 + 2\cancel{T} \times V_B = 0$

$4 - 5 + 2 \times V_B = 0$

$2V_B = 1$

$\therefore V_B = \frac{1}{2} m/s$, up↑